

Testing visibility diagnostics in AROME at high resolution

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1 Introduction

The main objective of this work has been to test and evaluate new visibility diagnostics fields coded in AROME at Météo France by Ingrid Etchevers (Dombrowski-Etchevers et al., 2018). In order to take full advantage of the physical parameterization of AROME, two configurations at very high resolution have been prepared: 1.3km and 0.5km resolution.

2 Visibility diagnostics in AROME CY46t1

Basically, visibility is defined as the distance up to which a meteorologist observer near the ground or the sea can see and identify an object in a given direction, at a specific time and place. This observation can be quantified and represented by a distance measure unit (meter, kilometer, feet, etc..). This parameter is reported regularly at the METAR observation messages (METeorological Aerodrome Report), and when it is significantly reduced, the obscuring weather phenomena is also reported (fog, haze, rain, dustrom, etc..).

The current study is based on visibility diagnostics coded at Météo France by Ingrid Etchevers (Dombrowski-Etchevers et al., 2018). This new diagnostic is calculated separately due to hydrometeors (precipitation) and clouds (fog) at the height of the lowest model level, which is 5 meters. However, this height could be defined at namelists by the key 'HVISI' :

- If HVISI < 5 meters, visibility will be calculated automatically at the lowest model level.
- If HVISI > 5 meters, visibility will be calculated at the predefined height at namelist.

The computation of the visibility is based on Koschmieder's formula which relates the horizontal visibility to the extinction coefficients of the different hydrometeors (liquid water, ice, snow, rain and graupel). The basic equations are the following :

$$PVISICLD = \frac{-\ln(0,05)}{0,013 + \beta_{cld} + \beta_{ice}} \quad (1)$$

Where :

PVISICLD : Visibility due to clouds (liquid water and ice water) in m;
 β_{cld} : Extinction coefficient due to cloud liquid water.
 β_{ice} : Extinction coefficient due to cloud ice water.

$$PVISIHVD = \frac{-\ln(0,05)}{0,013 + \beta_{rain} + \beta_{snow} + \beta_{graupel}} \quad (2)$$

Where :

PVISIHVD : Visibility due to hydrometeors in m;
 β_{rain} : Extinction coefficient due to rain.
 β_{ice} : Extinction coefficient due to snow.
 $\beta_{graupel}$: Extinction coefficient due to graupel.

Extinction coefficients β are calculated using the following relation: $\beta = a \cdot C^b$

Where :
 - C : Hydrometeors content (water - ice - rain - snow - graupel).
 - a, b : Coefficients determined empirically.

Those formulations are coded in the new routine "acvisih.F90" which is directly called from "apl_arome.F90", and are activated by switching-on the key LXVISI in the forecast namelist (LXVISI=.TRUE.). The next figure shows the main input required by this routine and the calculated outputs (PVISICLD, PVISIHYP and PMXCLWC).

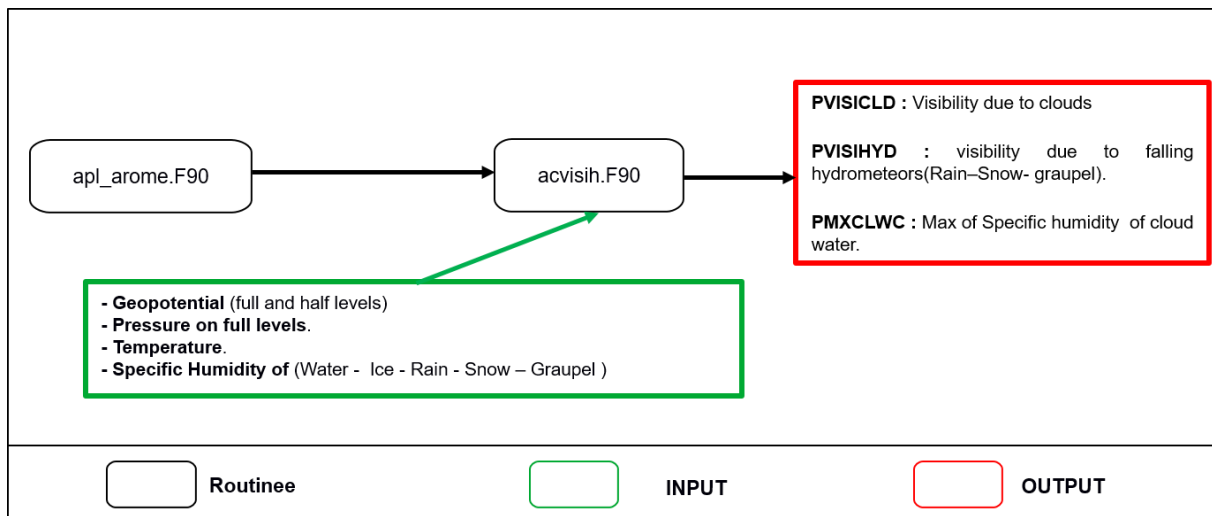


Figure : acvisih outputs and inputs organigram.

3 Experiment configuration

Study area

This study have been established on a small domain in Northern Algeria, covering the red area in figure 1. This region have been particularly picked because of the international airport of Algiers, where low visibility is frequently observed and thus it present a serious challenge for aeronautical forecasters due to its direct impact on airlines economics. The domain used for AROME 0.5km is smaller than the one used for AROME 1.3km just to reduce the computational costs of AROME 0.5km.

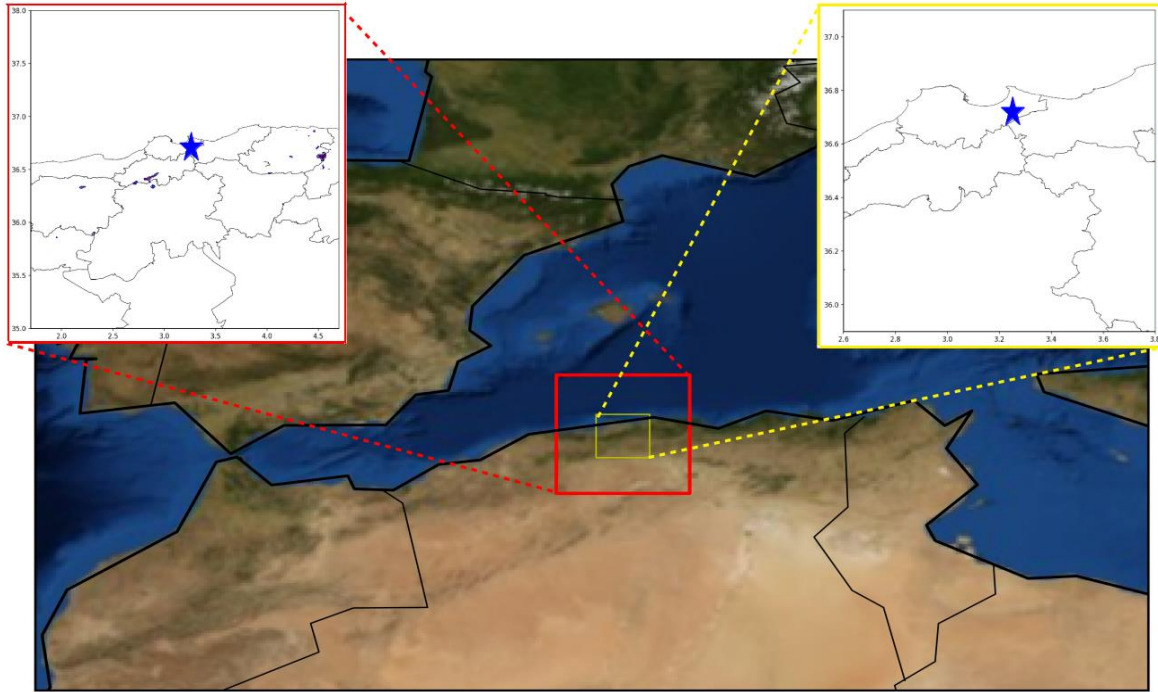


Figure 1: Study area location (red box : AROME 1.3 km domain, yellow box : AROME 0.5 km domain, blue star : international airport of Algiers).

Experiment setup

The occurrence of parameters reducing visibility (haze 1km to 5km, fog < 1 km) remain very sensitive to spatial resolution. So two configurations have been setup to evaluate the visibility diagnostics at different high resolution (1.3 and 0.5 km). We prepared two AROME configurations (table 1), both had 90 vertical levels based on CY46T1 and performed on the same study domain (figure 1).

Table 1: AROME configurations characteristics

Characteristics		AROME 1.3	AROME 0.5
Cycle		CY46T1.bf.06	
Initial conditions		ARPEGE	
Coupling frequency		1H	
Horizontal resolution		1.3 km	0.5 km
levels		90 level	
First level from ground		5 meters	
Time-step		40 s	10 s
Grid		232 x 232	241 x 241
Domain	Center	3.2°E – 36.5°N	
	Latitude	35°N – 38°N	35.9°N – 37.1°N
	Longitude	1.7°E – 4.7°E	2.6°E – 3.8°E

Simulations and evaluation

Simulations have been performed for two different months: December 2018 (28 day) and May 2018 (13 day). Observations data were obtained from METAR messages of the International airport station DAAG (36.71°N ; 3.21°E). Visibility data were organized into three categories:

- < 1000 m : very bad visibility conditions,
- [1000 m – 5000m[: bad visibility conditions,
- > 5000 m : good visibility conditions.

The evaluation will be based on **contingency tables** (table 2) in order to determine the model efficiency in predicting visibility conditions categories. This method is described as a best way of testing the skill especially in weather forecasts. The forecasting skill is evaluated by calculating various skill scores. From the contingency table, the following skill score will be calculated :

Table 2: Contingency table.

		Observation			Total
		Very bad	Bad	Good	
Forecast	Very bad	a	b	c	P
	Bad	d	e	f	O
	Good	g	h	i	N
	Total	J	K	L	M

- **Percentage Correct (PC)** : this score shows a fraction of the forecasts that is correct. It ranges from 0-1, with the perfect score being 1. This score can be misleading since it is heavily influenced by common category, usually no event in the case of rare events. It is determined by the formula :

$$PC = \frac{a + e + i}{M} * 100$$

- **Probability of detection/ hit rate (POD)** : POD shows usually the fraction of observed yes events that were correctly forecasted. The score ranges from 0-1, with score of 1 being the perfect score. It is sensitive to hits but ignores false alarm making it good for rare events. But for our case, operationally speaking, we'll consider that if the observed visibility is "Bad" and the forecast is "Very Bad", then the event is well predicted. The formula is :

$$PC = \frac{a + b + e + d}{J + K} * 100$$

- **False Alarm Ratio (FAR)** : It shows the percentage of the predicted yes events that actually did not occur. The FAR skill score ranges from 0-1 with the perfect score being 0. The skill score is sensitive to false alarms, but ignores misses. It is also very sensitive to the climatological frequency of the event. FAR skill score can be determined by applying the formula below :

$$PC = \frac{c + f}{P + O} * 100$$

Simulations have been performed for a total of 41 days : 28 days for December 2018 and 13 days for May 2018. The situations have been selected based on the METAR observations. For each situation, we performed 24h run starting from the day before the event (j-1).

Figures 2 and 3 shows the simulations of visibility due to clouds simulated by AROME 1.3km and AROME 0.5km at two different dates :

- On May 13th 2018 at 09h (figure 2), we notice in both configurations that the same area is concerned by low visibility values, few differences are detected inside the area where more details are provided in AROME 0.5km.

- On December 29th 2018 at 06h (figure 3), the differences are more important and the area concerned by the low visibility values is significantly more important in AROME 0.5km (red circle in figure 3).

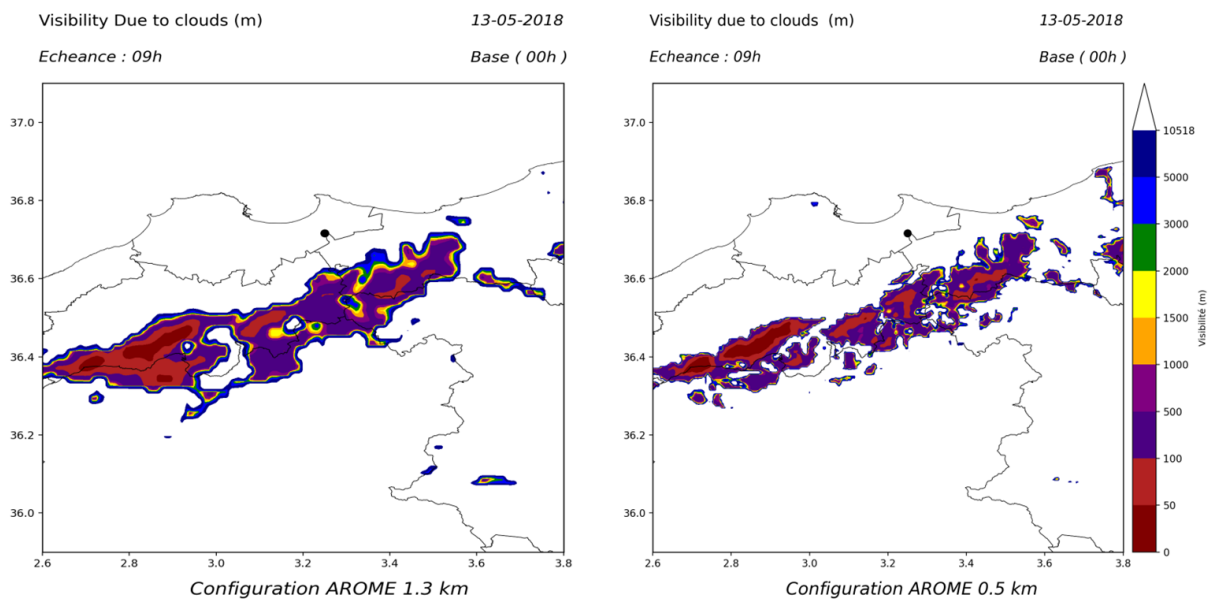


Figure 2: Comparison of visibility due to clouds simulated by AROME 1.3km and AROME 0.5km on 13 May 2018 at 09utc.

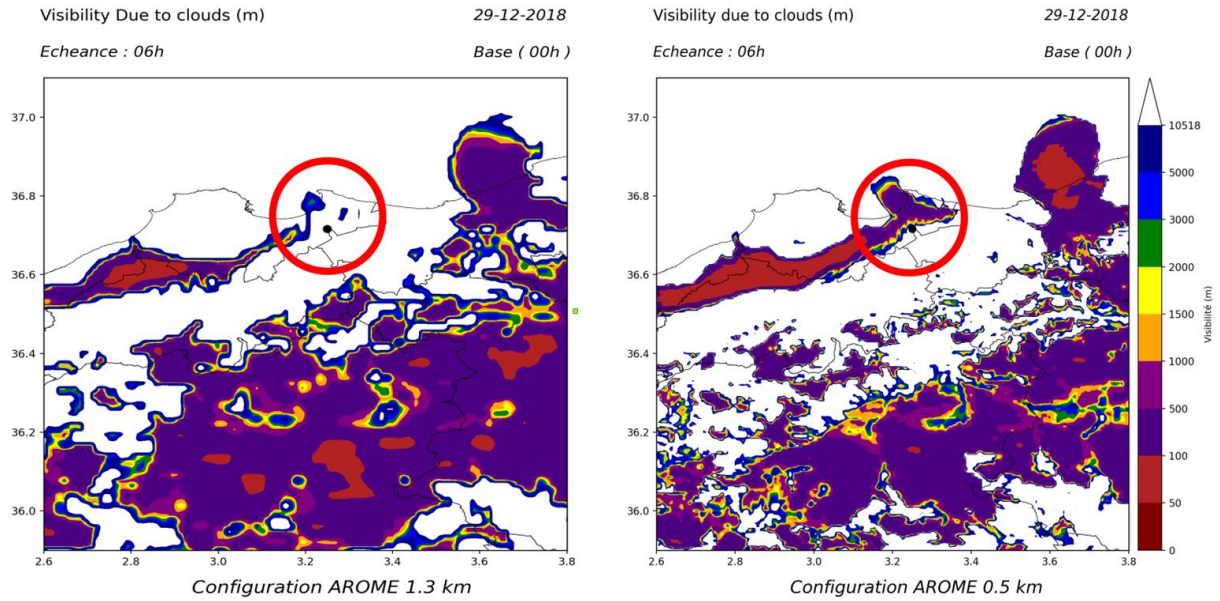


Figure 2: Comparison of visibility due to clouds simulated by AROME 1.3km and AROME 0.5km on 29 December 2018 at 06utc.

Table 3 and table 4 represents the contingency tables calculated based on hourly AROME 1.3km and AROME 0.5km simulations, respectively. Each one is decomposed into three parts : the whole period, December and May. Tab.05 represents the skill scores calculated for each configuration and for each period.

Table 3: Contingency table for AROME 1.3 km

		Observation (METAR)			
		All period			
Forecast (ARO 1.3)		VERY BAD	BAD	GOOD	Sum
	VERY BAD	17	40	46	103
	BAD	1	14	39	54
	GOOD	21	59	745	825
	Sum	39	113	830	982
	December				
		VERY BAD	BAD	GOOD	Sum
	VERY BAD	4	19	25	48
	BAD	1	1	22	24
	GOOD	16	43	539	598
	Sum	21	63	586	670
	May				
		VERY BAD	BAD	GOOD	Sum
	VERY BAD	13	21	21	55
	BAD	0	13	17	30
	GOOD	5	16	206	227
	Sum	18	50	244	312

Table 4: Contingency table for AROME 0.5 km

		Observation (METAR)			
		All period			
Forecast (ARO 0.5)		VERY BAD	BAD	GOOD	Sum
	VERY BAD	22	48	75	145
	BAD	2	10	28	40
	GOOD	15	55	727	797
	Sum	39	113	830	982
	December				
		VERY BAD	BAD	GOOD	Sum
	VERY BAD	8	24	45	77
	BAD	1	0	16	17
	GOOD	12	39	525	576
	Sum	21	63	586	670
	May				
		VERY BAD	BAD	GOOD	Sum
	VERY BAD	14	24	30	68
	BAD	1	10	12	23
	GOOD	3	16	202	221
	Sum	18	50	244	312

Table 5: Skill scores calculated for AROME 1.3 km and AROME 0.5 km.

	AROME 0.5	AROME 1.3
	All period	
PC	77,29	79,02
POD	0,53	0,47
FAR	0,55	0,54
	December	
PC	79,55	81,19
POD	0,38	0,29
FAR	0,64	0,65
	May	
PC	72,43	74,35
POD	0,71	0,69
FAR	0,46	0,44

The results are very promising for both configurations; the percentage of correct forecasts is quite high: 79% with AROME 1.3km and 77% with AROME 0.5km. In most cases, simulated visibility due to hydrometeors and clouds were in the same categories of the observed visibility from METAR.

The probability of detection of “very bad” and “bad” visibility classes is acceptable but not good enough for both resolutions: 0,47 for AROME 1.3km and 0,53 for AROME 0.5km. It shows that improving the horizontal resolution of AROME from 1.3 km to 0.5km, improved slightly the probability of low visibility conditions forecasts. In terms of seasons, we noticed that this skill score is significantly more important during May than during December. This should be proved by performing simulations for a longer period (a year at least).

The false alarm ratio score is just acceptable and quite similar for both configurations: 0,54 for AROME 1.3km and 0,55 for AROME 0.5km. It turns out that false alarms are more frequent during December than during May.

4 Conclusion

Two sets of simulations at different high horizontal resolution have been performed in this study in order to test new visibility diagnostics in AROME (Cy46T1). The evaluation of visibility due to clouds and hydrometeors forecasts showed some interesting results over the international airport of Algiers, with acceptable results regarding “very bad” and “bad” visibility conditions detection. It seems that the scores are slightly better in **May than in December**, but this still need more investigation by evaluating the forecasts at other airports and over a longer period.

5 References

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